

Attributing Probability to Language Models

Many people say that language models implement conditional probability distributions over next tokens. Are these people correct? I will argue no.

§1 a standard story about language model outputs

Many people say that¹

- (1) A language model is a function M that takes as input two arguments: a text token y and a string of text tokens x , and outputs a real number t ,
- (2) with the interpretation that t corresponds to the probability that the token that follows string x is y , i.e.

$$M(y \mid x) \text{ is proportional to } \Pr(y \text{ is next token} \mid x \text{ is string so far}).$$

where $\Pr(\cdot \mid \cdot)$ represents conditional probability, broadly construed.

§2 the issue with the standard story

2.1 the issue, informally

The problem with the standard story is the interpretation, i.e. part (2).

Mostly Responsible: At 4:58pm on July 24th, I have written all but the last paragraph of my AIDE paper (call this incomplete paper s). My moral scruples worn down, I put all this text into a large language model M and get outputs

$$M(\text{Language} \mid s) = 0.4, M(\text{In} \mid s) = 0.3, M(\text{John} \mid s) = 0.067, \dots$$

In *Mostly Responsible*, the standard story says $M(\text{Language} \mid s) = 0.4$ means the probability that “Language” is the next word to follow my incomplete essay s is 0.4, $M(\text{In} \mid s) = 0.3$ means the probability for “In” is 0.3, etc. But:

1. Is there really some thing, that can be more or less probable, that the model is measuring? i.e., what does it mean to say that “Language” is the next word to follow the string of my incomplete essay, especially if I haven’t written it?
2. Can we say that M could be more accurate or less accurate about the thing that can be more or less probable?

2.2 the issue, formally

This talk is about the following question, which I call LM-PROB:

Is it reasonable to attribute probabilities to the outputs of language models?

In general, I propose that in order for us to properly attribute probability to the output of a language model, the following criteria should be satisfied:²

1. *Relevance*: we take the output $M(y \mid x)$ to be relevant to the conditional probability $\Pr(Q_y \mid P_x)$ of some propositions P_x, Q_y , functions of x and y .
2. *Accuracy*: we evaluate the quality of the output $M(y \mid x)$ based on how accurate it is about how probable Q_y is given P_x .

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Handout for AIDE, July 27th, 2026
This handout: adrianliu.net/aidehandout
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¹ Here are some examples: Savcisen and Eliassi-Rad (2026):

Consider an LLM $\mathcal{M}$ with a vocabulary V . The LLM $\mathcal{M}$ maps the input text x to a probability distribution over subsequent tokens, denoted $P_{\mathcal{M}}$:

$$\mathcal{M}(x) = P_{\mathcal{M}}(\tau \mid x), \text{ where } \tau \in V.$$

For any token $\tau \in V$, the distribution $P_{\mathcal{M}}(\tau \mid x)$ denotes the probability that τ is the continuation of the sequence x .

Shanahan et al (2023):

the type of language model of interest here is a conditional probability distribution $P(w_{n+1} \mid w_1 \dots w_n)$, where $w_1 \dots w_n$ is a sequence of tokens (the context) and w_{n+1} is the predicted next token.

Keeling and Street (2026) write that the outputs of a large language model are:

probability distributions over tokens in the model’s vocabulary such that the probability assigned to each token is an estimate of how likely each token is to succeed the input sequence.

For good measure, OpenAI’s GPT 5.5 gives this definition for “large language model”:

a parameterized function $M_{\theta} : V^* \rightarrow \Delta(V)$ that maps a finite sequence of tokens from a vocabulary V to a probability distribution over the next token, thereby defining an autoregressive probability distribution over token sequences.

² *Relevance* is a version of a standard necessary condition for attributing representation in the philosophy of cognitive science, while *Accuracy* is a version of a standard necessary condition for attributing epistemic states in epistemology. My versions take inspiration from Harding (2023) and Hermann and Levinstein (2024). Another standard necessary condition for both is a “use” condition, specifying that the item in question (here, a language model output) has to play the right kind of role. I do *not* take a “use” condition to be necessary for attributing probability.

§3. attributing probability to language models

3.1 indicative conditional probabilities on training data. Language models are readily interpreted as *indicative* conditional next-token probabilities on training data.³ It is *prima facie* reasonable to think that we could just take this to be what is meant by “next token probabilities.”

TrainProb:⁴ $M(y \mid x)$ is proportional to the probability, if a masked string of the form $x[]$ is drawn from T , that the last token of $x[]$ is y .

TrainProb cannot be right in general. For inputs not drawn from T :

1. *relevance*: the premise that x is a masked element of the corpus T is always false. So $M(y \mid x)$ does not tell us anything nontrivial about strings drawn from outside the corpus T .⁵
2. *accuracy*: different candidate values of $M(y \mid x)$ cannot be considered more or less accurate, since there is no ground truth to compare them against.

3.2 indicative conditional probabilities on existing data. What if we appealed to the fact that strings drawn from outside training data are similar in relevant ways to strings drawn from the training data?⁶

IsProb: $M(y \mid x)$ is proportional to the probability, if $x[]$ is a masked string drawn from a target domain similar to the training data, that the last token is y .

IsProb cannot be right in general. For inputs nobody has composed before:

1. *relevance*: the premise that x is a masked element of some set X of existing strings is always false. So $M(y \mid x)$ does not tell us anything nontrivial about strings that have never been written.
2. *accuracy*: different candidate values of $M(y \mid x)$ cannot be considered more or less accurate, since there is no ground truth to compare them against.⁷

3.3 counterfactual conditional probabilities. What if instead we understand the output of M as indicating the probability that x *would be* followed by y if x were to be generated by some data-generating process?

WouldProb:⁸ $M(y \mid x)$ is proportional to the probability, if data generating process $\mathcal{D}$ were to output masked string $x[]$, that the last token of x would be y .

If we could specify what the process $\mathcal{D}$ is, then *relevance* and *accuracy* would be satisfied.⁹ But there are metaphysical and epistemic hurdles to specifying $\mathcal{D}$:

1. *metaphysical*: a reference class-style problem. Any string x can be coherently understood as drawn from different data generating processes.¹⁰ What makes any particular $\mathcal{D}$ the “right” distribution to consider x an output from? And why would it be of interest to model the process $\mathcal{D}$?
2. *epistemic*: for any choice of underlying data-generating process $\mathcal{D}$, do we have good reason to think that the model successfully models *that* process? I think WouldProb is the only strategy that has any hope of giving an affirmative answer to LM-PROB. But it faces real hurdles.

³ This is roughly because the training process of a language model is designed to minimize the distance between model outputs and some “ground-truth” about next tokens across some training set T .

⁴ Here a masked element $x[]$ is a string of tokens that comprises the substring x plus one last token at the end, which is hidden from view.

⁵ When models are fine-tuned, the notion of a corpus of “training data” gets harder to specify. I think the situation only gets worse here. Ask me about this!

⁶ A motivating thought: this is how classifier models generalize outside their training data. Perhaps language models could do the same?

⁷ Here is the disanalogy with classifiers. Even when data goes beyond training data, we only expect good results from the classifier on inputs where there is a ground truth about what label is correct.

⁸ How much would would a WouldProb prob if a WouldProb would prob would? inspired by E.L. Riggsbee.

⁹ The relevant probabilities would be the ones specified by WouldProb, according to $\mathcal{D}$, and accuracy of the model M could be measured relative to the true probabilities of the process $\mathcal{D}$.

¹⁰ Indeed, for a corpus of textual training data, the following all seem to have valid claims to be a process that generated the data:

- (a) The process of humans generating all and only the content in the corpus.
- (b) The process of humans generating content between 1500 and 2026 CE.
- (c) The process of humans generating content.

§4. frequently-asked questions

4.1 how about credence and belief?

Q: Others¹¹ have discussed whether we can attribute epistemic states, like credence and belief, to language models. Is LM-PROB different? What is new?

A: It is related but distinct. My question differs because (1) epistemic states don't necessarily have to be probabilistic and (2) probabilistic structures don't have to be epistemic states.¹² But the questions are related in two ways:

1. Credences *are* probabilistic, and criteria like *accuracy* and *relevance* likely do apply to them.¹³ In this case, any attribution of credence would need an interpretation of the structures that satisfies *accuracy* and *relevance*.
2. If some structures played the *functional role* of belief or credence within a language model architecture, but did *not* satisfy *accuracy* and *relevance*, these would be bad epistemic states that we should not be taking seriously!

4.2 does this only apply to the output?

Q: Recent work in interpretability research has used probes to attempt to extract semantically-significant data from internal architectures implementing language models. If probes can extract probability-like content from middle layers, could those be called probabilities?

A: The worries of §3 rearise here. E.g. suppose a probe is able to extract values in internal layers of a language model that correspond to probabilities of some events over a corpus. When we input novel strings into the language model, what probabilities would the probe-extracted values represent?¹⁴

4.3 do we have to think of language model outputs as probabilities?

Q: Do we have to think of language model outputs as probabilities?

A: Shanahan et al (2023) suggest a “simulation” interpretation of some language model outputs.¹⁵ They still take the outputs to literally be probabilities,¹⁶ but we could instead think in terms of a non-probabilistic “proposal strength”:

SimProp: $M(y \mid x)$ is proportional to the strength of the proposal that y be the token following string x , relative to other proposals $M(\cdot \mid x)$.

We can then understand the training of a model as favoring string continuations that lead the string to resemble the training data in particular ways. This non-probabilistic interpretation allows us to sidestep the *accuracy* and *relevance* criteria. But the theory of probability would need to be replaced by a similarly robust theory of similarity, and questions like the ones arising for WouldProb would prob rearise. e.g.

1. What are the dimensions of similarity that the model gives and how does the real-valued output capture these?
2. What is the significance of the particular dimensions of similarity to the training data that the model gives?

← i.e. likely to be output when sampling i.i.d. from a data-generating distribution $\mathcal{D}$ that generates interesting and relevant AIDE participant-style questions to the paper.

¹¹ e.g. Harding (2023), Hermann and Levinstein (2024), Hernandez Flowerman (2026), Keeling and Street (2026)

¹² This is why I don't include a *use* criterion: I think we can understand some output as a probability as observers if *we* can use it as a probability, even if the system itself does not use it that way.

¹³ Hermann and Levinstein (2025) argue this

¹⁴ Some truth-tracking probes, like Azaria and Mitchell (2023), Hermann and Levinstein (2025), Savcicens and Eliassi-Rad (2026), restrict the setting to inputs that are always true or false propositions. This basically treats the language model as a classifier.

¹⁵ Metaphorically, different textual continuations can implement different possible characters for the textual output to “role-play”, and the model's distribution over next tokens can be seen as a superposition of different characters, or “simulacra”, that the textual output can instantiate.

¹⁶ see quote on p.1 of this handout

§5. actually-asked questions

My wonderful commentator Meredith poses the following questions:

1: verbal dispute?

M: *To what extent is the debate over whether to interpret language models as outputting conditional probability distributions over next tokens a **verbal dispute**?*

2: degrees of reasonableness?

M: *To **what extent** is it unreasonable to understand the output of a generative model M as indicating the probability that x would be followed by y if x were to be generated by some data-generating process?*

3: shouldn't simulations be relevant and accurate too?

M: *Is the **simulation interpretation** subject to analogous relevance and accuracy conditions [as the probability interpretations]?*

Selected References

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